

Career Operations Intelligence Platform

Prompt Engineering Specification

Overview

The platform relies on multiple AI-assisted decision points rather than a single monolithic prompt.

This architecture emerged from practical experimentation during development.

Early prototypes attempted to combine extraction, evaluation, ranking, and recommendation tasks into a single inference step. While this reduced workflow complexity, it consistently produced lower-quality results. Models frequently mixed extraction with evaluation, introduced unsupported assumptions, lost source information, and generated recommendations that were difficult to trace back to original inputs.

The platform was subsequently redesigned around task separation.

Each AI subsystem now performs a narrowly defined responsibility with explicit constraints, structured outputs, and deterministic validation layers.

This document describes the purpose, design rationale, and operational boundaries of each prompt.

Design Principles

Several prompt-engineering principles guide the platform.

Single Responsibility

Each model invocation should perform one primary task.

Extraction prompts should extract.

Evaluation prompts should evaluate.

Routing prompts should route.

Combining multiple responsibilities within a single prompt increases hallucination risk and reduces output consistency.

Structured Outputs

All production prompts return structured JSON.

Structured outputs reduce ambiguity, simplify validation, and enable deterministic downstream processing.

Whenever possible, free-form narrative generation is avoided.

Evidence-Based Evaluation

Models are instructed to evaluate only information explicitly present within source material.

Unsupported assumptions should reduce confidence rather than generate conclusions.

Human Review

No prompt is authorized to make final application decisions.

All outputs are advisory and subject to human review.

Workflow A

Pass 1

Opportunity Extraction Engine

Purpose

Pass 1 identifies employment opportunities contained within incoming emails and converts them into structured records.

The objective is extraction rather than evaluation.

Input

Normalized email dataset.

Output

Structured opportunity records.

Example:

```
{  
  "title": "",  
  "company": "",  
  "location": "",  
  "candidate_urls": [],  
  "confidence": 0.0  
}
```

Explicit Constraints

Pass 1 may:

- ❖ identify jobs
- ❖ split digest emails
- ❖ extract URLs
- ❖ preserve source metadata

Pass 1 may not:

- ❖ rank jobs
- ❖ recommend applications
- ❖ evaluate fit
- ❖ estimate attainability
- ❖ recommend resumes

Lessons Learned

Early versions attempted to combine extraction and ranking.

This frequently caused opportunities to be omitted because the model prematurely judged them as low quality.

Separating extraction from evaluation improved recall and reduced information loss.

Pass 2

Career Intelligence Evaluator

Purpose

Pass 2 evaluates extracted opportunities and determines which opportunities deserve attention.

Input

Canonical job dataset produced by Pass 1.

Output

Ranked opportunity recommendations.

Optimization Targets

The prompt is optimized for:

- ❖ interview probability
- ❖ career capital
- ❖ long-term optionality
- ❖ organizational development value
- ❖ geographic practicality
- ❖ compensation quality

Explicit Constraints

Pass 2 may:

- ❖ evaluate
- ❖ compare
- ❖ rank
- ❖ prioritize

Pass 2 may not:

- ❖ invent requirements
- ❖ invent compensation
- ❖ invent responsibilities
- ❖ infer qualifications not present
- ❖ modify source data

Lessons Learned

Without explicit constraints, the model frequently filled information gaps using assumptions.

This produced recommendations that appeared reasonable but could not be traced back to source material.

Prompt revisions increasingly emphasized uncertainty handling and confidence reduction when information was incomplete.

Workflow B

Resume Routing Engine

Purpose

Determine which resume profile appears most appropriate for a selected opportunity.

Input

Job description.

Resume metadata profiles.

Output

Resume recommendation.

Terminology analysis.

Confidence assessment.

Explicit Constraints

The prompt may:

- ❖ recommend resumes
- ❖ identify terminology gaps
- ❖ compare skills inventories

The prompt may not:

- ❖ rewrite resumes
- ❖ generate experience
- ❖ fabricate qualifications
- ❖ create resume content

Lessons Learned

Initial versions frequently drifted into resume-writing behavior.

Additional constraints were introduced to maintain focus on routing and terminology analysis rather than content generation.

Reliability Controls

Several controls were introduced in response to observed model behavior.

Structured JSON Enforcement

All prompts return machine-readable outputs.

Canonicalization Layers

Source information is normalized before evaluation.

Confidence Scoring

Models are required to express uncertainty explicitly.

Human Review Gates

No recommendation is acted upon automatically.

Deterministic Validation

Outputs are validated before entering downstream workflows.

Observed Failure Modes

The most common failure modes encountered during development included:

- ❖ unsupported assumptions
- ❖ role seniority misclassification
- ❖ omission of source information
- ❖ ranking drift
- ❖ inconsistent JSON formatting
- ❖ overconfidence in incomplete data

Many architectural decisions within the platform were introduced specifically to mitigate these behaviors.

Key Findings

The project demonstrated that large language models perform best when used as bounded reasoning systems rather than autonomous workflow engines.

Performance improved substantially when:

- ❖ responsibilities were separated
- ❖ prompts were narrowly scoped
- ❖ outputs were structured
- ❖ deterministic controls were introduced
- ❖ human review remained in the loop

The resulting architecture treats AI as a decision-support component within a larger system rather than the system itself.